Practical 3 - Implementing the Servlet IO and File applications.
Theory –
MultipartConfig
fileSizeThreshold
location
maxFileSize
maxRequestSize
servletOutputStream class
ReadListener Interface
WriteListener Interface
3a. Create a Servlet application to upload and download a file.

i) Servlet application to upload file

Index.htnl

```
<html>
<form action="FileUploadServlet" method="post" enctype="multipart/form-data">
   <h1>File Uploading</h1>
Select File to Upload:<input type="file" name="file"><br>
Destination <input type="text"
value="C:\Users\ITCS\Documents\NetBeansProjects\Demo3_a\src\main\webapp\tmp"
name="destination">
<br>
<input type="submit" value="FileUpload" name="upload">
</form>
</html>
```

FileUploadServlet.java

```
package fileservletapp;
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.annotation.MultipartConfig;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;
@WebServlet("/FileUploadServlet")
@MultipartConfig
public class FileUploadServlet extends HttpServlet {
@Override
public void doPost(HttpServletRequest req,HttpServletResponse res) throws
ServletException, IOException
{
res.setContentType("text/html");
PrintWriter out = res.getWriter();
String path=req.getParameter("destination");
out.println(path);
```

```
Part filePart=req.getPart("file");
String sfilePart=req.getPart("file").toString();
out.print("<br> filePart: "+sfilePart);
String filename=filePart.getSubmittedFileName();
out.print("<br><br><hr> file name: "+filename);
OutputStream os=null;
InputStream is=null;
try {
os=new FileOutputStream(new File(path+File.separator+filename));
is=filePart.getInputStream();
int read=0;
byte[] b=new byte[1024];
while ((read = is.read(b)) != -1) {
os.write(b, 0, read);
}
out.println("<br>file uploaded sucessfully...!!!");
out.close();
}
catch(FileNotFoundException e){out.print(e);}
} }
```

ii) Servlet application to download file

index.html

```
<html>
<head>
<title>File Download Page</title>
</head>
<body>
<h1>File Download Application</h1>
Click <a href="DownloadServlet?filename=Request Dispatcher.pdf">Request
Dispatcher</a>
<br/><br/>
Click <a href="DownloadServlet?filename=Cookies.pdf">Cookies</a>
</body>
</html>
```

~~~~~ DownloadServlet.java ~~~~~~

```
package filedownloadapp;
import java.io.IOException;
import java.io.InputStream;
import java.io.PrintWriter;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.*;
@WebServlet("/DownloadServlet")
```
~~~~~

```
public class DownloadServlet extends HttpServlet {
@Override
public void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
response.setContentType("APPLICATION/OCTET-STREAM");
String filename = request.getParameter("filename");
ServletContext context = getServletContext();
InputStream is = context.getResourceAsStream("/" + filename);
//ServletOutputStream out = response.getOutputStream(); // any of the two works
PrintWriter out=response.getWriter();
response.setHeader("Content-Disposition","attachment; filename=\"" + filename + "\"");
//if comment this statement then it wl ask you about the editor with which you want to open
the file
int i;
while ((i=is.read()) != -1) {
out.write(i);
}
is.close();
out.close();
}
}
```

3b. Develop Simple Servlet Question Answer Application using Database.

In MySql command line client do the following

a) Open your database using USE command
b) Give the following commands

```
create table quiz (qnovarchar(5) PRIMARY KEY,questionvarchar(100), op1
varchar(50), op2 varchar(50), op3 varchar(50), op4 varchar(50), ansvarchar(50));

insert into quiz values('001','What is the capital of India??','New
Delhi','Kolkata','Chennai','Mumbai',"1");

insert into quiz values('002','Who was the First President of India??','Dr. Rajendra
Prasad','Dr. S. Radhakrishnan','RamNathKovind','V. V. Giri','1');

insert into quiz values('003','What is ORM','Object Ratio Mean','Object Rotation
Measure','Object Relation Mapping','Oracle Request Management','3');

insert into quiz values('004','Unit of Energy is ___','Dozon','Kilo Meter
','Joul','Hertz','3');

insert into quiz values('005',' --- is the smallest memory
unit.','bit','byte','KiloByte','GigaByte','1');
```

Index.html

```
<html><head><title>Quiz Application</title></head>
<body>
<h1>Welcome to Quiz Servlet </h1>
<h1><a href="QuizServlet" >CLICK TO START QUIZ</a></h1>
</body>
</html>
```

QuizServlet.java

```
package MyPack;

import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.sql.*;

@WebServlet(name = "QuizServlet", urlPatterns = {"/QuizServlet"})
public class QuizServlet extends HttpServlet {
```

```
@Override
public void doGet(HttpServletRequest request, HttpServletResponse response)throws
ServletException, IOException {
response.setContentType("text/html;charset=UTF-8");
PrintWriter out = response.getWriter();
out.println("<form method=post action=ShowResult>");
try {
Class.forName("com.mysql.cj.jdbc.Driver");
Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/logindb?autoReconnect=
true&useSSL=false","root","tiger");
Statement stmt = con.createStatement();
ResultSet res = stmt.executeQuery("select * from quiz");
out.println("<table border=1 >");
int qno=0;
while(res.next()){
qno++;
out.println("<tr><td>"+res.getString(1)+"</td>");
out.println("<td>"+res.getString(2)+"</td></tr>");
out.println("<tr><td><input type=radio name="+qno+"
value=1>"+"</td><td>"+res.getString(3)+"</td></tr>");
out.println("<tr><td><input type=radio name="+qno+"
value=2>"+"</td><td>"+res.getString(4)+"</td></tr>");
out.println("<tr><td><input type=radio name="+qno+"
value=3>"+"</td><td>"+res.getString(5)+"</td></tr>");
out.println("<tr><td><input type=radio name="+qno+"
value=4>"+"</td><td>"+res.getString(6)+"</td></tr>");
}
}catch(ClassNotFoundException | SQLException e)
{out.println(e);}
out.println("</table>");
out.println("<input type=reset >");
out.println("<input type=submit value=SUBMIT >");
out.println("</form>");
}
}
```

ShowResult.java

```
packageMyPack;

import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
```

```
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.*;
import java.sql.*;
@WebServlet(name = "ShowResult", urlPatterns = {"/ShowResult"})
public class ShowResult extends HttpServlet {

public void doGet(HttpServletRequest request, HttpServletResponse response)
throwsServletException, IOException {
response.setContentType("text/html;charset=UTF-8");
PrintWriter out = response.getWriter();
try {
Class.forName("com.mysql.cj.jdbc.Driver");
Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/quizdb?autoReconnect=t
rue&useSSL=false","root","tiger");
Statement stmt = con.createStatement();
ResultSet res = stmt.executeQuery("select ans from quiz");
int count =0, qno=0;
while(res.next()){
if(res.getString(1).equals(request.getParameter(""+(++qno))))
{ count++;
out.println("<h1>Correct </h1>");
}
else {
out.println("<h1>Incorrect </h1>");
}
}
out.println("<h1>Your Score is "+count+" </h1>");
}catch(Exception e){out.println(e);}}}
```

3c. Create simple Servlet application to demonstrate Non-Blocking Read Operation.

Index.html

```
<html>
<head>
<title>Non Blocking IO</title>
<meta charset="UTF-8">
<meta http-equiv="Refresh" content="0; URL=NonBlockingServlet">
</head>
<body>
</body>
</html>
```

NonBlockingServlet.java

```
packageMyPack;

import java.io.*;
import java.net.*;
import java.net.HttpURLConnection;
import java.net.URL;
import java.util.logging.Level;
import java.util.logging.Logger;
import jakarta.servlet.*;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;
@WebServlet(name = "NonBlockingServlet", urlPatterns = {"/NonBlockingServlet"})
public class NonBlockingServlet extends HttpServlet {
protected void service(HttpServletRequest request, HttpServletResponse response)
throwsServletException, IOException {
response.setContentType("text/html;charset=UTF-8");
try (PrintWriter out = response.getWriter()) {

out.println("<h1>FileReader</h1>");
 String filename="/WEB-INF/BookList.txt";
ServletContext c=getServletContext();
InputStream in=c.getResourceAsStream(filename);
 String path;
path =
"http://"+request.getServerName()+":"+request.getServerPort()+request.getContextPath()+"/
ReadingNonBloclingServlet";
 URL url=new URL(path);
HttpURLConnection conn=(HttpURLConnection)url.openConnection();
conn.setChunkedStreamingMode(2);
conn.setDoOutput(true);
conn.connect();
```

```
if(in!=null)
 {
InputStreamReaderinr=new InputStreamReader(in);
BufferedReaderbr = new BufferedReader(inr);
 String text="";
System.out.println("Reading started....");
BufferedWriter bw=new BufferedWriter(new
OutputStreamWriter(conn.getOutputStream()));
while((text=br.readLine())!=null){
out.print(text+"<br>");
try{
Thread.sleep(1000);
out.flush();
 }
catch(InterruptedExceptionex){}
}out.print("reading completed....");
bw.flush();
bw.close();
 }
 }
 }
}
```

ReadingNonBlockingServlet.java

```
import java.io.*;
import jakarta.servlet.AsyncContext;
import jakarta.servlet.ServletException;
import jakarta.servlet.ServletInputStream;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet(urlPatterns = {"/ReadingNonBlockingServlet"},asyncSupported = true )
public class ReadingNonBlockingServlet extends HttpServlet {
@Override
protected void service(HttpServletRequest request, HttpServletResponse response)
throwsServletException, IOException
{ response.setContentType("text/html");
AsyncContext ac = request.startAsync();
ServletInputStream in=request.getInputStream();
in.setReadListener(new ReadingListener(in,ac));
}
}
```

ReadingListener.java

```
packageMyPack;
```

```
import jakarta.servlet.AsyncContext;
import jakarta.servlet.ServletInputStream;
import jakarta.servlet.ReadListener;
public class ReadingListener implements ReadListener
{ ServletInputStream input = null;
AsyncContext ac = null;
ReadingListener(ServletInputStream in, AsyncContext c) {
input = in;
ac = c;
}
@Override
public void onDataAvailable() {
}
public void onAllDataRead()
{ ac.complete();
}
public void onError(Throwable t)
{ ac.complete();
t.printStackTrace();}}
```